AME 5101 - Applied Linear Algebra

Assignment 2

Maximum marks: 10 Duration: 45 minutes

Embedding and Layer Norm - A Worked Example.

Position Embedding and Layer Normalisation [CO2-PO3-BL4]

A model has $d_e = 4$, with token- and positional-embedding matrices

$$\mathbf{W}_e = \begin{pmatrix} 1 & 0 & 2 & 1 & 0 \\ 0 & 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 1 & 1 \\ 1 & 2 & 0 & 3 & 0 \end{pmatrix}, \qquad \mathbf{W}_p = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix},$$

where vocabulary indices run from 1 to 5 and positions from 1 to 3.

(a) **[5]** Demonstrate how the input to the first layer is formed for token $v = 2$ at position $t = 3$: identify the column of $\mathbf{W}_e$ and the column of $\mathbf{W}_p$ that are used, and determine the embedding $\mathbf{e} = \mathbf{W}_e[:, v] + \mathbf{W}_p[:, t]$.

Answer. The token embedding is column $v = 2$ of $\mathbf{W}_e$ and the positional embedding is column $t = 3$ of $\mathbf{W}_p$ (columns are tokens; indices start at 1): $\mathbf{W}_e[:, 2] = (0,\ 2,\ 1,\ 2)^\top$ and $\mathbf{W}_p[:, 3] = (1,\ 1,\ 0,\ 1)^\top$. Adding them, $\mathbf{e} = (1,\ 3,\ 1,\ 3)^\top$.

Marking: [2] the two correct columns (1 each); [3] the embedding $\mathbf{e}$.

(b) **[5]** Demonstrate the effect of LayerNorm, with $\boldsymbol{\gamma} = \mathbf{1}$ and $\boldsymbol{\beta} = \mathbf{0}$, on the embedding $\mathbf{e}$ from part (a): determine the mean m, the variance $\sigma^2 = \frac{1}{d_e}\sum_i (e_i - m)^2$ and the normalised vector $(\mathbf{e} - m)/\sigma$, and deduce the mean and variance of the normalised vector.

Answer. Mean $m = (1+3+1+3)/4 = 8/4 = 2$. Deviations $\mathbf{e} - m = (-1,\ 1,\ -1,\ 1)^\top$. Variance $\sigma^2 = ((-1)^2 + (1)^2 + (-1)^2 + (1)^2)/4 = 4/4 = 1$, so $\sigma = 1$. With $\boldsymbol{\gamma} = \mathbf{1}$, $\boldsymbol{\beta} = \mathbf{0}$ the output is $(\mathbf{e} - m)/\sigma = (-1,\ 1,\ -1,\ 1)^\top$. Its entries sum to 0 (mean 0) and their squares average to 1 (variance 1): LayerNorm removes the offset and the scale of the column.

Marking: [1] mean; [2] variance; [1] normalised vector; [1] mean 0 and variance 1. If $\mathbf{e}$ from (a) is wrong, award (b) on method.

The Assignment Question.

Q1. [CO2-PO3-BL4] **10 Marks**

Answer the following question on the same lines as the worked example shown above. A model has $d_e = 4$, with token- and positional-embedding matrices

$$\mathbf{W}_e = \begin{pmatrix} 2 & 0 & 1 & 2 & 1 \\ 1 & 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 3 & 1 \\ 1 & 0 & 2 & 1 & 0 \end{pmatrix}, \qquad \mathbf{W}_p = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{pmatrix},$$

where vocabulary indices run from 1 to 5 and positions from 1 to 3.

(a) [5] Examine how the input to the first layer is formed for token $v = 4$ at position $t = 2$: identify the column of $\mathbf{W}_e$ and the column of $\mathbf{W}_p$ that are used, and determine the embedding $\mathbf{e} = \mathbf{W}_e[:, v] + \mathbf{W}_p[:, t]$.

(b) [5] Apply the effect of LayerNorm, with $\boldsymbol{\gamma} = \mathbf{1}$ and $\boldsymbol{\beta} = \mathbf{0}$, on the embedding $\mathbf{e}$ from part (a): determine the mean m, the variance $\sigma^2 = \frac{1}{d_e}\sum_i (e_i - m)^2$ and the normalised vector $(\mathbf{e} - m)/\sigma$, and deduce the mean and variance of the normalised vector.